Ch06_varselect_lab

April 16, 2026

[1]:
```
from functools import partial
print = partial(print, flush=True)
```

[2]:
```
import numpy as np
import pandas as pd
from matplotlib.pyplot import subplots
from statsmodels.api import OLS
import sklearn.model_selection as skm
import sklearn.linear_model as skl
from sklearn.preprocessing import StandardScaler
from ISLP import load_data
from ISLP.models import ModelSpec as MS
from functools import partial
```

[3]:
```
from sklearn.pipeline import Pipeline
from sklearn.decomposition import PCA
from sklearn.cross_decomposition import PLSRegression
from ISLP.models import \
     (Stepwise,
      sklearn_selected,
      sklearn_selection_path)
from l0bnb import fit_path
```

[4]:
```
import warnings
warnings.simplefilter("ignore")
```

[5]:
```
Hitters = load_data('Hitters')
np.isnan(Hitters['Salary']).sum()
```

[5]: 59

[6]:
```
Hitters = Hitters.dropna();
Hitters.shape
```

[6]: (263, 20)

[7]:
```
def nCp(sigma2, estimator, X, Y):
    "Negative Cp statistic"
```

```
    n, p = X.shape
    Yhat = estimator.predict(X)
    RSS = np.sum((Y - Yhat)**2)
    return -(RSS + 2 * p * sigma2) / n
```

[8]:
```
design = MS(Hitters.columns.drop('Salary')).fit(Hitters)
Y = np.array(Hitters['Salary'])
X = design.transform(Hitters)
sigma2 = OLS(Y,X).fit().scale
```

[9]:
```
neg_Cp = partial(nCp, sigma2)
```

[10]:
```
strategy = Stepwise.first_peak(design, direction='forward',␣
↪max_terms=len(design.terms))
```

[11]:
```
hitters_MSE = sklearn_selected(OLS, strategy, scoring='neg_mean_squared_error')
hitters_MSE.fit(Hitters, Y)
hitters_MSE.selected_state_
```

[11]:
```
('Assists',
 'AtBat',
 'CAtBat',
 'CHits',
 'CHmRun',
 'CRBI',
 'CRuns',
 'CWalks',
 'Division',
 'Errors',
 'Hits',
 'HmRun',
 'League',
 'NewLeague',
 'PutOuts',
 'RBI',
 'Runs',
 'Walks',
 'Years')
```

[12]:
```
hitters_Cp = sklearn_selected(OLS,
                              strategy,
                              scoring=neg_Cp)
hitters_Cp.fit(Hitters, Y)
hitters_Cp.selected_state_
```

[12]:
```
('Assists',
 'AtBat',
```

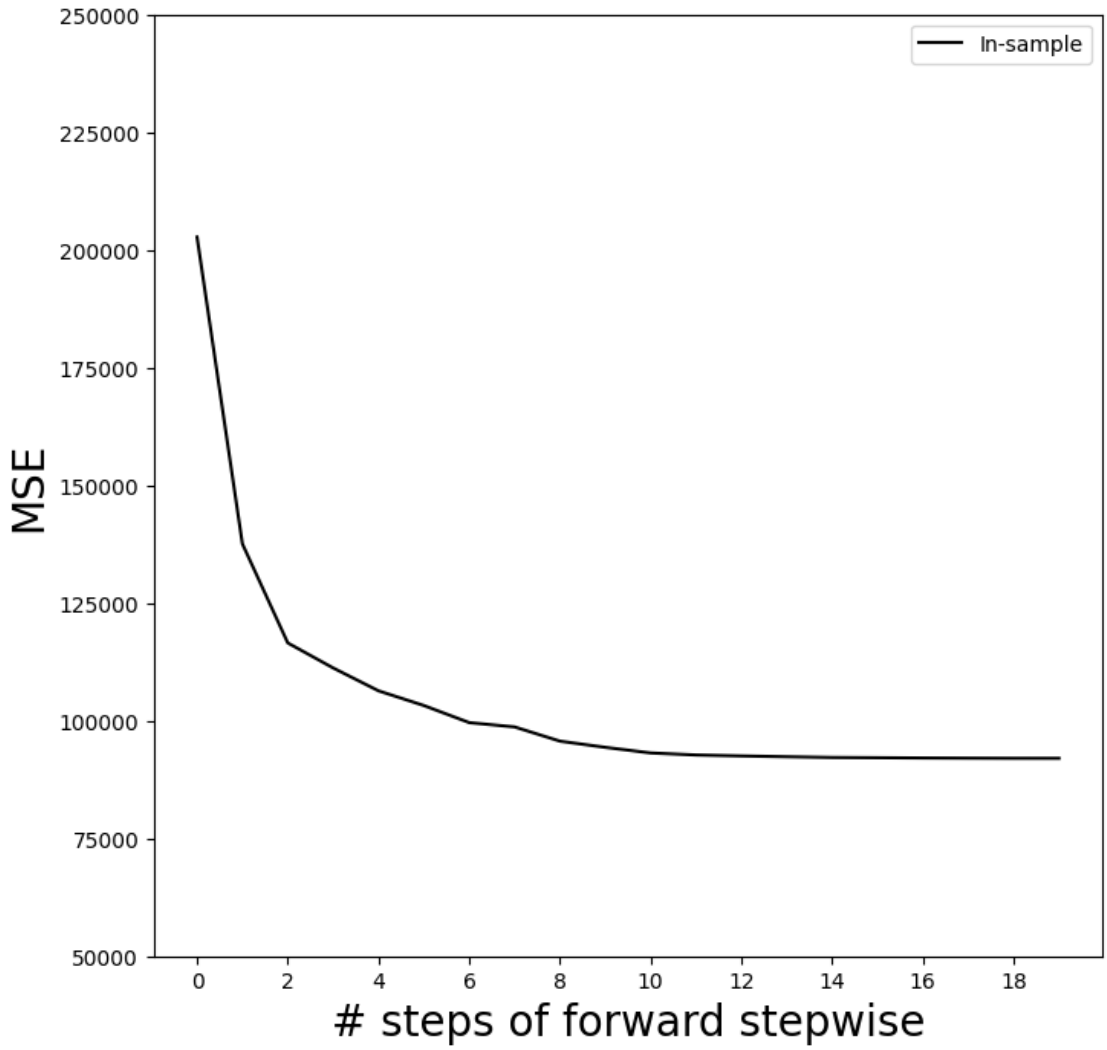
```
 'CAtBat',
 'CRBI',
 'CRuns',
 'CWalks',
 'Division',
 'Hits',
 'PutOuts',
 'Walks')
```

[13]:
```
strategy = Stepwise.fixed_steps(design, len(design.terms), direction='forward')
full_path = sklearn_selection_path(OLS, strategy,␣
 ↪scoring='neg_mean_squared_error')
```

[14]:
```
full_path.fit(Hitters, Y)
Yhat_in = full_path.predict(Hitters)
Yhat_in.shape
```

[14]: (263, 20)

[15]:
```
mse_fig, ax = subplots(figsize=(8,8))
insample_mse = ((Yhat_in - Y[:,None])**2).mean(0)
n_steps = insample_mse.shape[0]
ax.plot(np.arange(n_steps),
        insample_mse,
        'k', # color black
        label='In-sample')
ax.set_ylabel('MSE',
              fontsize=20)
ax.set_xlabel('# steps of forward stepwise',
              fontsize=20)
ax.set_xticks(np.arange(n_steps)[::2])
ax.legend()
ax.set_ylim([50000,250000]);
```



[16]:
```
K = 5
kfold = skm.KFold(K, random_state=0, shuffle=True)
Yhat_cv = skm.cross_val_predict(full_path, Hitters, Y, cv=kfold)
Yhat_cv.shape
```

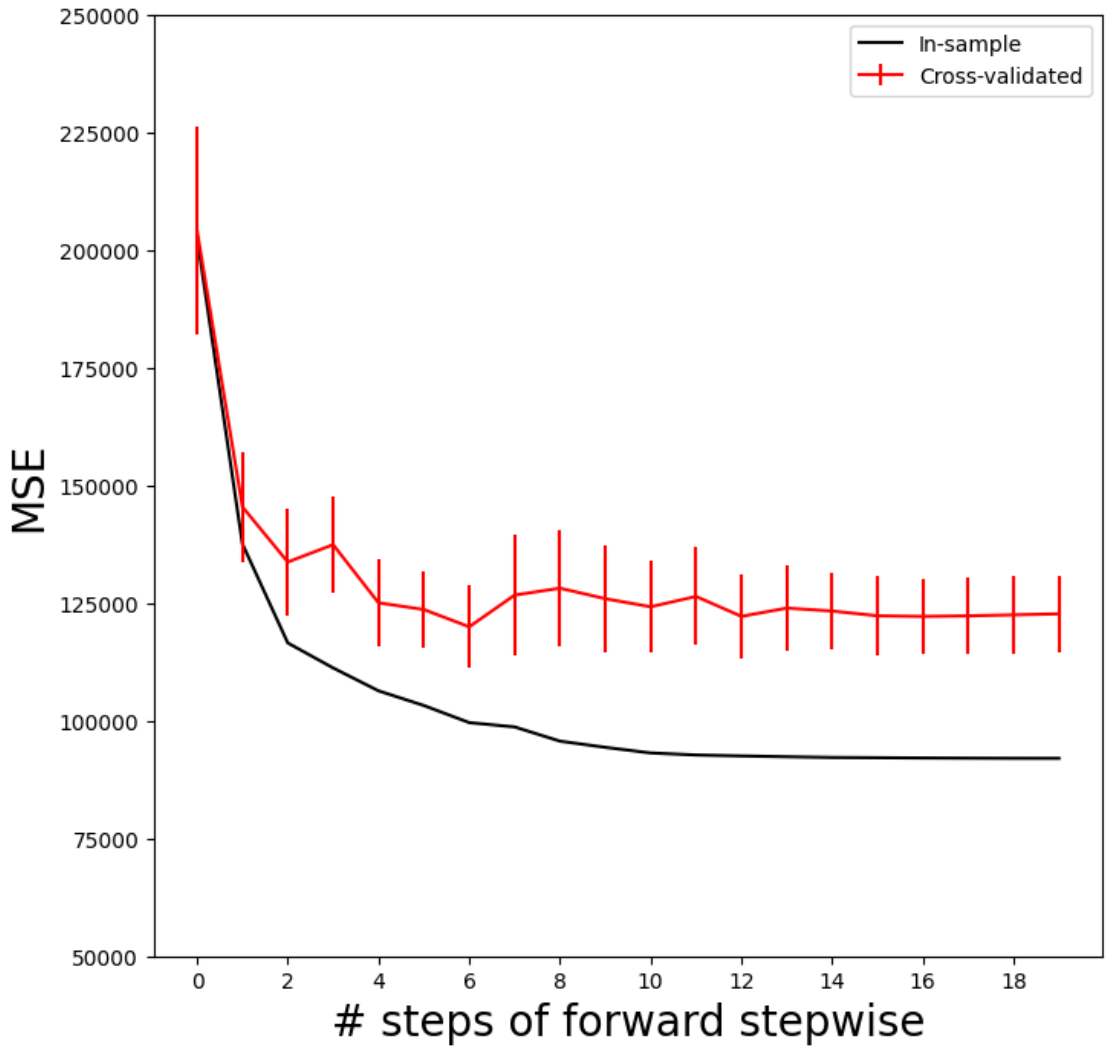
[16]: (263, 20)

[17]:
```
cv_mse = []
for train_idx, test_idx in kfold.split(Y):
    errors = (Yhat_cv[test_idx] - Y[test_idx,None])**2
    cv_mse.append(errors.mean(0)) # column means
cv_mse = np.array(cv_mse).T
cv_mse.shape
```

[17]: (20, 5)

[18]:
```
ax.errorbar(np.arange(n_steps),
            cv_mse.mean(1),
            cv_mse.std(1) / np.sqrt(K),
            label='Cross-validated',
            c='r') # color red
ax.set_ylim([50000,250000])
ax.legend()
mse_fig
```

[18]:

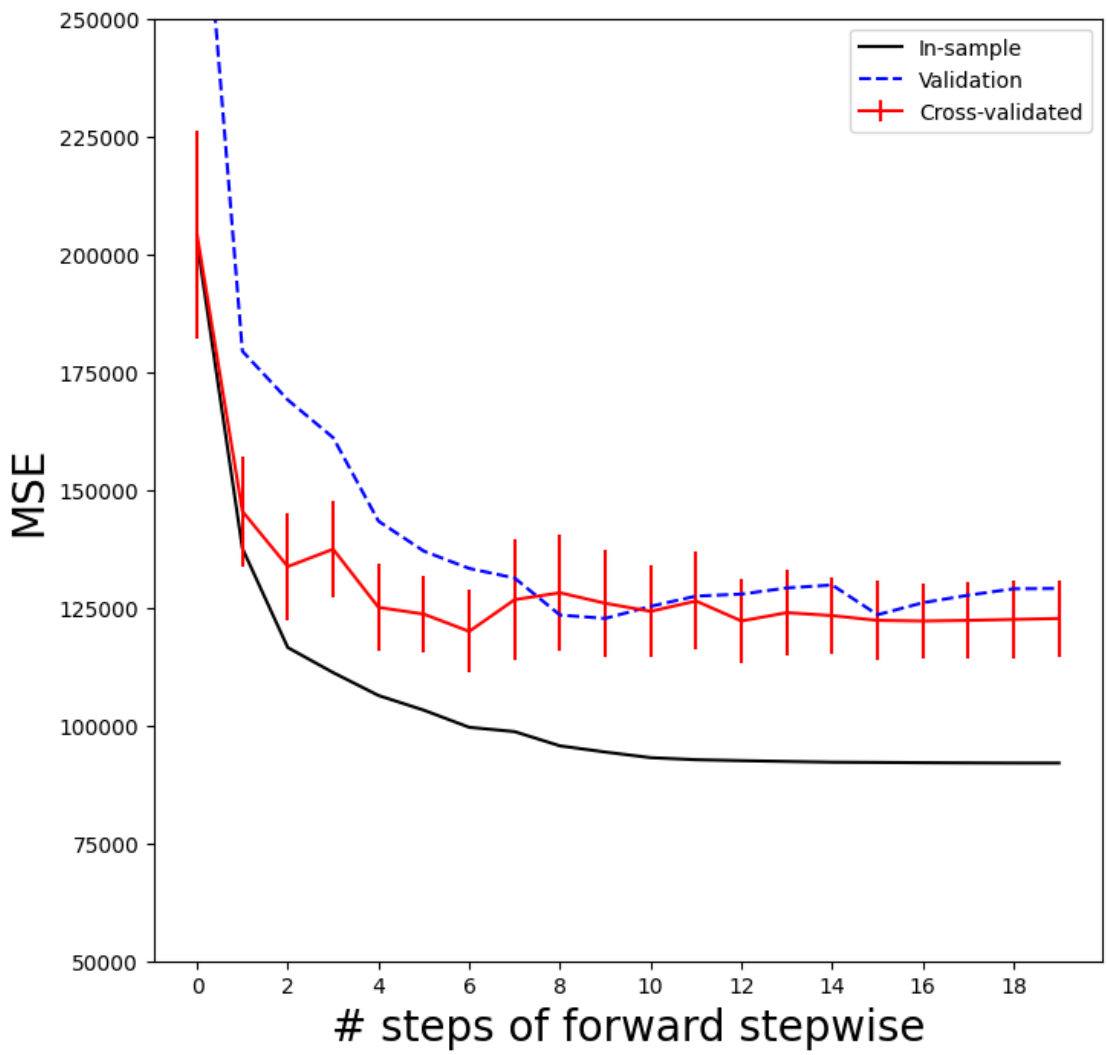


[19]:
```
validation = skm.ShuffleSplit(n_splits=1,
                              test_size=0.2,
                              random_state=0)
```

```
for train_idx, test_idx in validation.split(Y):
    full_path.fit(Hitters.iloc[train_idx],
                  Y[train_idx])
    Yhat_val = full_path.predict(Hitters.iloc[test_idx])
    errors = (Yhat_val - Y[test_idx,None])**2
    validation_mse = errors.mean(0)
```

[20]:
```
ax.plot(np.arange(n_steps),
        validation_mse,
        'b--', # color blue, broken line
        label='Validation')
ax.set_xticks(np.arange(n_steps)[::2])
ax.set_ylim([50000,250000])
ax.legend()
mse_fig
```

[20]:



[21]:
```
D = design.fit_transform(Hitters)
D = D.drop('intercept', axis=1)
X = np.asarray(D)
```

[22]:
```
path = fit_path(X, Y, max_nonzeros=X.shape[1])
```

```
Preprocessing Data.
BnB Started.
Iteration: 1. Number of non-zeros:  1
Iteration: 2. Number of non-zeros:  2
Iteration: 3. Number of non-zeros:  2
Iteration: 4. Number of non-zeros:  2
Iteration: 5. Number of non-zeros:  3
Iteration: 6. Number of non-zeros:  3
Iteration: 7. Number of non-zeros:  4
Iteration: 8. Number of non-zeros:  9
Iteration: 9. Number of non-zeros:  9
Iteration: 10. Number of non-zeros:  9
Iteration: 11. Number of non-zeros:  9
Iteration: 12. Number of non-zeros:  9
Iteration: 13. Number of non-zeros:  9
Iteration: 14. Number of non-zeros:  9
Iteration: 15. Number of non-zeros:  9
Iteration: 16. Number of non-zeros:  9
Iteration: 17. Number of non-zeros:  9
Iteration: 18. Number of non-zeros:  17
Iteration: 19. Number of non-zeros:  19
```

[23]:
```
path[3]
```

[23]:
```
{'B': array([0.        , 3.25484367, 0.        , 0.        , 0.        ,
        0.        , 0.        , 0.        , 0.        , 0.        ,
        0.        , 0.67775265, 0.        , 0.        , 0.        ,
        0.        , 0.        , 0.        , 0.        ]),
 'B0': -38.98216739555505,
 'lambda_0': 0.011416248027450187,
 'M': 0.5829861733382012,
 'Time_exceeded': False}
```

[24]:
```
Xs = X - X.mean(0)[None,:]
X_scale = X.std(0)
Xs = Xs / X_scale[None,:]
lambdas = 10**np.linspace(8, -2, 100) / Y.std()
soln_array = skl.ElasticNet.path(Xs,
                                 Y,
                                 l1_ratio=0.,
                                 alphas=lambdas)[1]
soln_array.shape
```

[24]: (19, 100)

[25]:
```
soln_path = pd.DataFrame(soln_array.T, columns=D.columns, index=-np.
 ↪log(lambdas))
soln_path.index.name = 'negative log(lambda)'
soln_path
```

[25]:
```
                          AtBat        Hits      HmRun       Runs        RBI  \
negative log(lambda)
-12.310855             0.000800    0.000889   0.000695   0.000851   0.000911
-12.078271             0.001010    0.001122   0.000878   0.001074   0.001150
-11.845686             0.001274    0.001416   0.001107   0.001355   0.001451
-11.613102             0.001608    0.001787   0.001397   0.001710   0.001831
-11.380518             0.002029    0.002255   0.001763   0.002158   0.002310
…                           …           …          …          …          …
 9.784658           -290.823989  336.929968  37.322686 -59.748520 -26.507086
 10.017243          -290.879272  337.113713  37.431373 -59.916820 -26.606957
 10.249827          -290.923382  337.260446  37.518064 -60.051166 -26.686604
 10.482412          -290.958537  337.377455  37.587122 -60.158256 -26.750044
 10.714996          -290.986528  337.470648  37.642077 -60.243522 -26.800522

                          Walks      Years      CAtBat      CHits     CHmRun  \
negative log(lambda)
-12.310855             0.000900   0.000812    0.001067   0.001113   0.001064
-12.078271             0.001135   0.001025    0.001346   0.001404   0.001343
-11.845686             0.001433   0.001293    0.001698   0.001772   0.001694
-11.613102             0.001808   0.001632    0.002143   0.002236   0.002138
-11.380518             0.002281   0.002059    0.002704   0.002821   0.002698
…                           …          …           …          …          …
 9.784658            134.855915 -17.216195 -387.775826  89.573601 -12.273926
 10.017243           134.900549 -17.108041 -388.458404  89.000707 -12.661459
 10.249827           134.936136 -17.022194 -388.997470  88.537380 -12.971603
 10.482412           134.964477 -16.954081 -389.423414  88.164178 -13.219329
 10.714996           134.987027 -16.900054 -389.760135  87.864551 -13.416889

                          CRuns        CRBI      CWalks  League[N]  \
negative log(lambda)
-12.310855             0.001141    0.001149    0.000993  -0.000029
-12.078271             0.001439    0.001450    0.001253  -0.000037
-11.845686             0.001816    0.001830    0.001581  -0.000046
-11.613102             0.002292    0.002309    0.001995  -0.000058
-11.380518             0.002892    0.002914    0.002517  -0.000073
…                           …           …           …          …
 9.784658            476.079273  257.271255 -213.124780  31.258215
 10.017243           477.031349  257.966790 -213.280891  31.256434
 10.249827           477.791860  258.523025 -213.405740  31.254958
 10.482412           478.398404  258.967059 -213.505412  31.253747
```

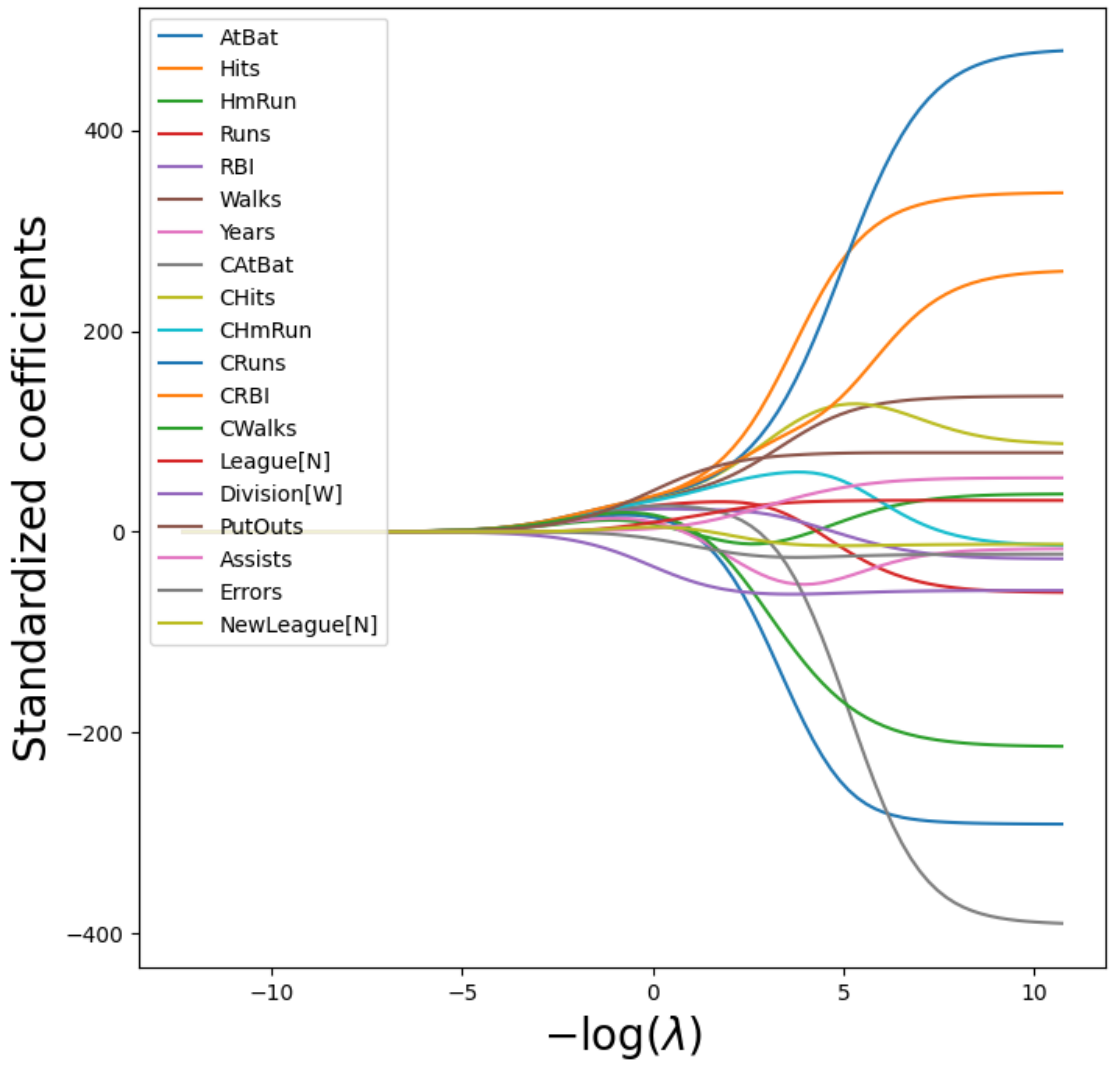
```
 10.714996            478.881540  259.321007 -213.584869  31.252760

                      Division[W]    PutOuts    Assists     Errors  \
negative log(lambda)
-12.310855              -0.000390   0.000609   0.000052  -0.000011
-12.078271              -0.000492   0.000769   0.000065  -0.000014
-11.845686              -0.000621   0.000970   0.000082  -0.000017
-11.613102              -0.000784   0.001224   0.000104  -0.000022
-11.380518              -0.000990   0.001544   0.000131  -0.000028
…                            …          …          …         …
 9.784658              -58.457857  78.761266  53.622113 -22.208456
 10.017243             -58.448850  78.761240  53.645147 -22.198802
 10.249827             -58.441682  78.761230  53.663357 -22.191071
 10.482412             -58.435983  78.761230  53.677759 -22.184893
 10.714996             -58.431454  78.761235  53.689152 -22.179964

                      NewLeague[N]
negative log(lambda)
-12.310855               -0.000006
-12.078271               -0.000007
-11.845686               -0.000009
-11.613102               -0.000012
-11.380518               -0.000015
…                             …
 9.784658               -12.402891
 10.017243              -12.391969
 10.249827              -12.383205
 10.482412              -12.376191
 10.714996              -12.370587

[100 rows x 19 columns]
```

[26]:
```
path_fig, ax = subplots(figsize=(8,8))
soln_path.plot(ax=ax, legend=False)
ax.set_xlabel('$-\log(\lambda)$', fontsize=20)
ax.set_ylabel('Standardized coefficients', fontsize=20)
ax.legend(loc='upper left');
```



[27]:
```
beta_hat = soln_path.loc[soln_path.index[39]]
lambdas[39], beta_hat
```

[27]:
```
(25.53538897200662,
 AtBat           5.433750
 Hits            6.223582
 HmRun           4.585498
 Runs            5.880855
 RBI             6.195921
 Walks           6.277975
 Years           5.299767
 CAtBat          7.147501
 CHits           7.539495
 CHmRun          7.182344
```

```
CRuns           7.728649
CRBI            7.790702
CWalks          6.592901
League[N]       0.042445
Division[W]    -3.107159
PutOuts         4.605263
Assists         0.378371
Errors         -0.135196
NewLeague[N]    0.150323
Name: -3.240065292879872, dtype: float64)
```

[28]:
```
np.linalg.norm(beta_hat)
```

[28]:
```
24.17061720144378
```

[29]:
```
beta_hat = soln_path.loc[soln_path.index[59]]
lambdas[59], np.linalg.norm(beta_hat)
```

[29]:
```
(0.24374766133488554, 160.42371017725918)
```

[30]:
```
ridge = skl.ElasticNet(alpha=lambdas[59], l1_ratio=0)
scaler = StandardScaler(with_mean=True,  with_std=True)
pipe = Pipeline(steps=[('scaler', scaler), ('ridge', ridge)])
pipe.fit(X, Y)
```

[30]:
```
Pipeline(steps=[('scaler', StandardScaler()),
                ('ridge', ElasticNet(alpha=0.24374766133488554, l1_ratio=0))])
```

[31]:
```
np.linalg.norm(ridge.coef_)
```

[31]:
```
160.42371017725927
```

[32]:
```
validation = skm.ShuffleSplit(n_splits=1,
                              test_size=0.5,
                              random_state=0)
ridge.alpha = 0.01
results = skm.cross_validate(ridge,
                             X,
                             Y,
                             scoring='neg_mean_squared_error',
                             cv=validation)
-results['test_score']
```

[32]:
```
array([134214.00419204])
```

[33]:
```
ridge.alpha = 1e10
results = skm.cross_validate(ridge,
                             X,
```

```
                              Y,
                              scoring='neg_mean_squared_error',
                              cv=validation)
-results['test_score']
```

[33]: array([231788.32155285])

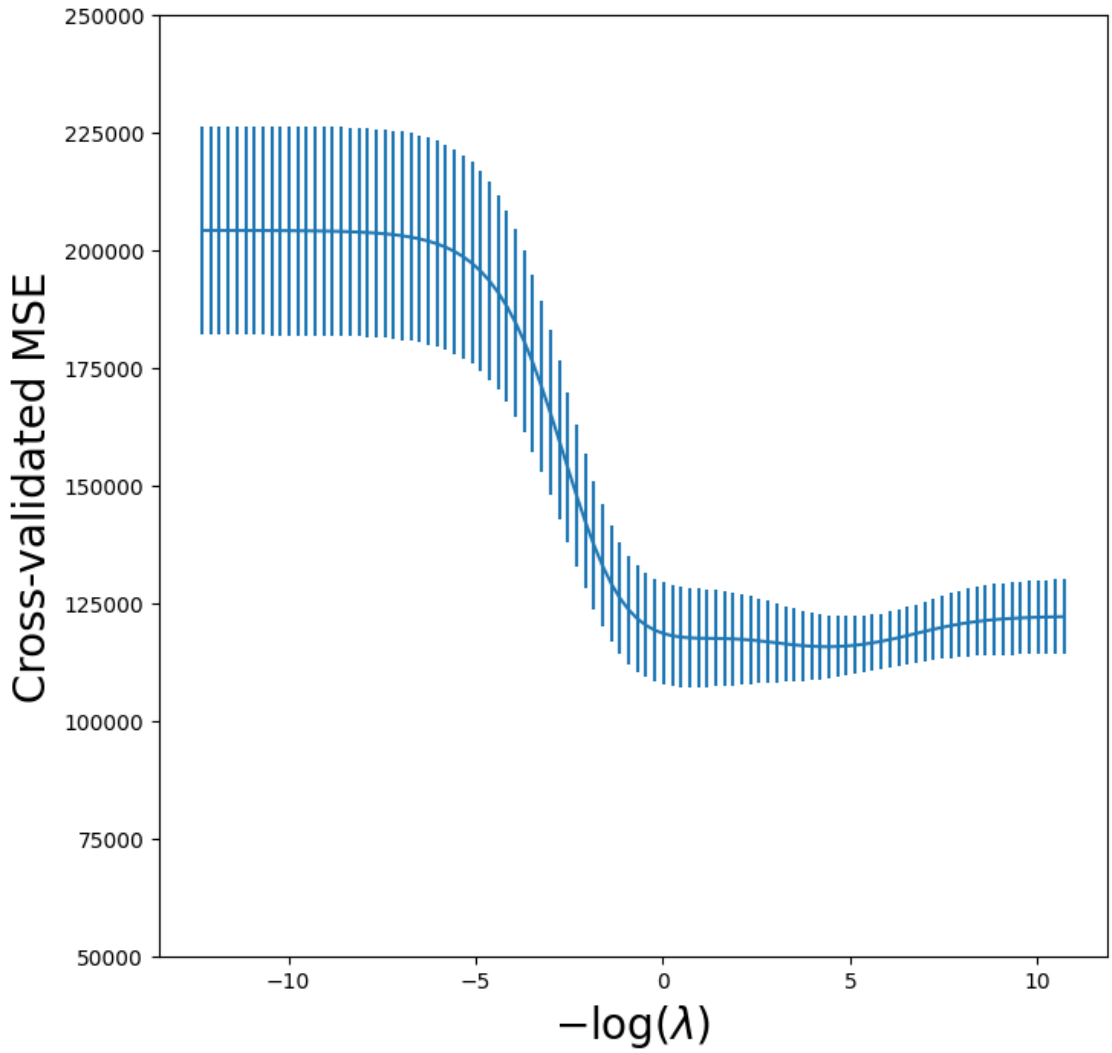
[34]:
```
param_grid = {'ridge__alpha': lambdas}
grid = skm.GridSearchCV(pipe,
                        param_grid,
                        cv=validation,
                        scoring='neg_mean_squared_error')
grid.fit(X, Y)
grid.best_params_['ridge__alpha']
grid.best_estimator_
```

[34]:
```
Pipeline(steps=[('scaler', StandardScaler()),
                ('ridge', ElasticNet(alpha=0.005899006046740856, l1_ratio=0))])
```

[35]:
```
grid = skm.GridSearchCV(pipe,
                        param_grid,
                        cv=kfold,
                        scoring='neg_mean_squared_error')
grid.fit(X, Y)
grid.best_params_['ridge__alpha']
grid.best_estimator_
```

[35]:
```
Pipeline(steps=[('scaler', StandardScaler()),
                ('ridge', ElasticNet(alpha=0.01185247763144249, l1_ratio=0))])
```

[36]:
```
ridge_fig, ax = subplots(figsize=(8,8))
ax.errorbar(-np.log(lambdas),
            -grid.cv_results_['mean_test_score'],
            yerr=grid.cv_results_['std_test_score'] / np.sqrt(K))
ax.set_ylim([50000,250000])
ax.set_xlabel('$-\log(\lambda)$', fontsize=20)
ax.set_ylabel('Cross-validated MSE', fontsize=20);
```

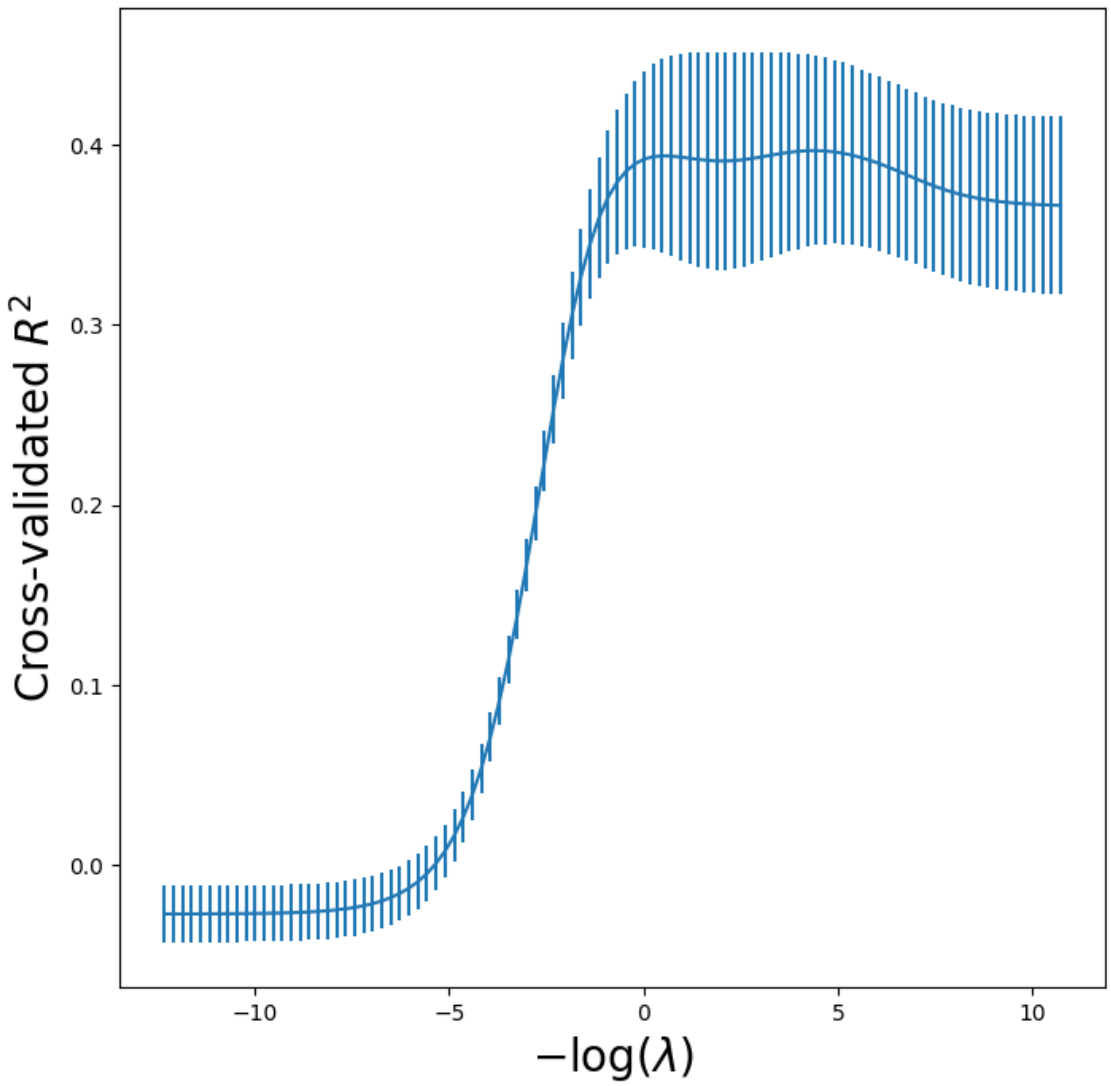


[37]:
```
grid_r2 = skm.GridSearchCV(pipe,
                           param_grid,
                           cv=kfold)
grid_r2.fit(X, Y)
```

[37]:
```
GridSearchCV(cv=KFold(n_splits=5, random_state=0, shuffle=True),
             estimator=Pipeline(steps=[('scaler', StandardScaler()),
                                       ('ridge',
                                        ElasticNet(alpha=10000000000.0,
                                                   l1_ratio=0))]),
             param_grid={'ridge__alpha': array([2.22093791e+05, 1.76005531e+05,
1.39481373e+05, 1.10536603e+05,
       8.75983676e+04, 6.94202082e+04, 5.50143278e+04, 4.35979140e+04,
       3.45506012e+04, 2.73807606…
```

```
4.67486141e-03, 3.70474772e-03, 2.93594921e-03, 2.32668954e-03,
1.84386167e-03, 1.46122884e-03, 1.15799887e-03, 9.17694298e-04,
7.27257037e-04, 5.76338765e-04, 4.56738615e-04, 3.61957541e-04,
2.86845161e-04, 2.27319885e-04, 1.80147121e-04, 1.42763513e-04,
1.13137642e-04, 8.96596467e-05, 7.10537367e-05, 5.63088712e-05,
4.46238174e-05, 3.53636122e-05, 2.80250579e-05, 2.22093791e-05])})
```

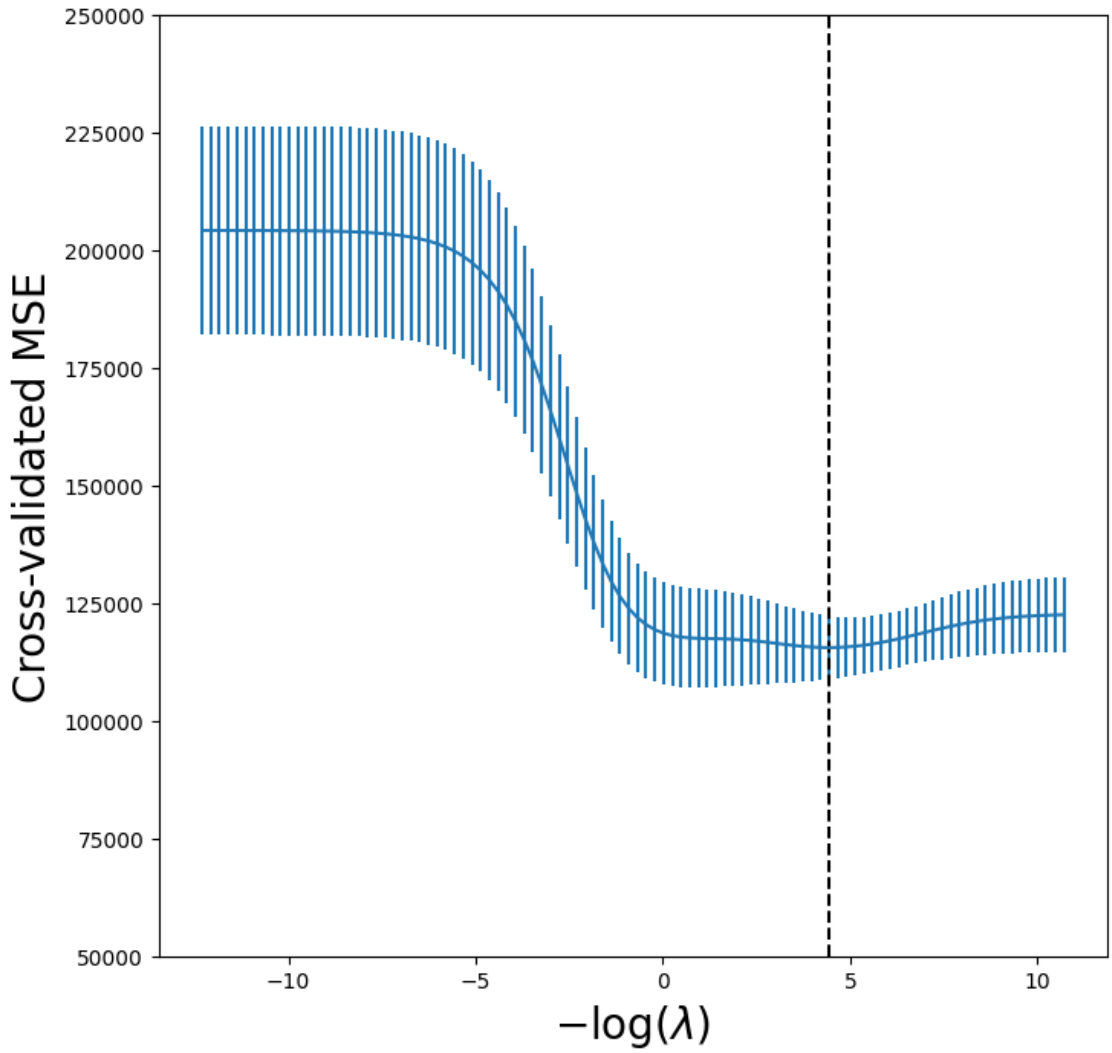
[38]:
```
r2_fig, ax = subplots(figsize=(8,8))
ax.errorbar(-np.log(lambdas),
            grid_r2.cv_results_['mean_test_score'],
            yerr=grid_r2.cv_results_['std_test_score'] / np.sqrt(K))
ax.set_xlabel('$-\log(\lambda)$', fontsize=20)
ax.set_ylabel('Cross-validated $R^2$', fontsize=20);
```



[39]:
```
ridgeCV = skl.ElasticNetCV(alphas=lambdas,
                           l1_ratio=0,
                           cv=kfold)
pipeCV = Pipeline(steps=[('scaler', scaler),
                         ('ridge', ridgeCV)])
pipeCV.fit(X, Y)
```

[39]:
```
Pipeline(steps=[('scaler', StandardScaler()),
                ('ridge',
                 ElasticNetCV(alphas=array([2.22093791e+05, 1.76005531e+05,
1.39481373e+05, 1.10536603e+05,
       8.75983676e+04, 6.94202082e+04, 5.50143278e+04, 4.35979140e+04,
       3.45506012e+04, 2.73807606e+04, 2.16987845e+04, 1.71959156e+04,
       1.36274691e+04, 1.07995362e+04, 8.55844774e+03, 6.78242347e+03,
       5.37495461e+03, 4.25955961e+03,…
       1.84386167e-03, 1.46122884e-03, 1.15799887e-03, 9.17694298e-04,
       7.27257037e-04, 5.76338765e-04, 4.56738615e-04, 3.61957541e-04,
       2.86845161e-04, 2.27319885e-04, 1.80147121e-04, 1.42763513e-04,
       1.13137642e-04, 8.96596467e-05, 7.10537367e-05, 5.63088712e-05,
       4.46238174e-05, 3.53636122e-05, 2.80250579e-05, 2.22093791e-05]),
                              cv=KFold(n_splits=5, random_state=0,
shuffle=True),
                              l1_ratio=0))])
```

[40]:
```
tuned_ridge = pipeCV.named_steps['ridge']
ridgeCV_fig, ax = subplots(figsize=(8,8))
ax.errorbar(-np.log(lambdas),
            tuned_ridge.mse_path_.mean(1),
            yerr=tuned_ridge.mse_path_.std(1) / np.sqrt(K))
ax.axvline(-np.log(tuned_ridge.alpha_), c='k', ls='--')
ax.set_ylim([50000,250000])
ax.set_xlabel('$-\log(\lambda)$', fontsize=20)
ax.set_ylabel('Cross-validated MSE', fontsize=20);
```



[41]:
```
np.min(tuned_ridge.mse_path_.mean(1))
```

[41]: 115526.70630987812

[42]:
```
tuned_ridge.coef_
```

[42]:
```
array([-222.80877051,  238.77246614,    3.21103754,   -2.93050845,
          3.64888723,  108.90953869,  -50.81896152, -105.15731984,
        122.00714801,   57.1859509 ,  210.35170348,  118.05683748,
       -150.21959435,   30.36634231,  -61.62459095,   77.73832472,
         40.07350744,  -25.02151514,  -13.68429544])
```

[43]:
```
outer_valid = skm.ShuffleSplit(n_splits=1,
                               test_size=0.25,
```

```
                                random_state=1)
inner_cv = skm.KFold(n_splits=5,
                     shuffle=True,
                     random_state=2)
ridgeCV = skl.ElasticNetCV(alphas=lambdas,
                           l1_ratio=0,
                           cv=inner_cv)
pipeCV = Pipeline(steps=[('scaler', scaler),
                         ('ridge', ridgeCV)]);
```

[44]:
```
results = skm.cross_validate(pipeCV,
                             X,
                             Y,
                             cv=outer_valid,
                             scoring='neg_mean_squared_error')
-results['test_score']
```

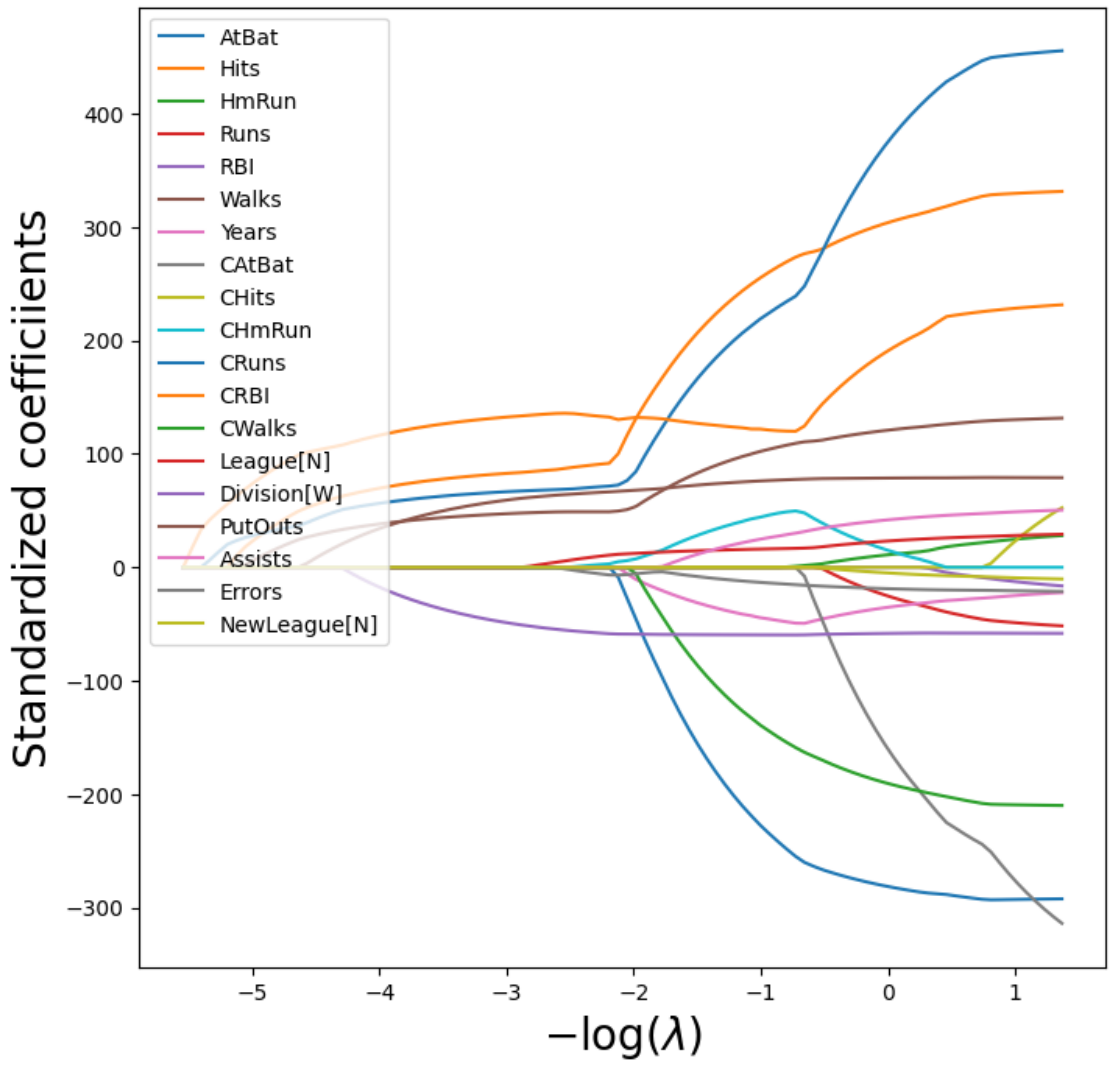
[44]: array([132393.84003227])

[45]:
```
lassoCV = skl.ElasticNetCV(n_alphas=100,
                           l1_ratio=1,
                           cv=kfold)
pipeCV = Pipeline(steps=[('scaler', scaler),
                         ('lasso', lassoCV)])
pipeCV.fit(X, Y)
tuned_lasso = pipeCV.named_steps['lasso']
tuned_lasso.alpha_
```

[45]: 3.1472370031649866

[46]:
```
lambdas, soln_array = skl.Lasso.path(Xs,
                                     Y,
                                     l1_ratio=1,
                                     n_alphas=100)[:2]
soln_path = pd.DataFrame(soln_array.T,
                         columns=D.columns,
                         index=-np.log(lambdas))
```

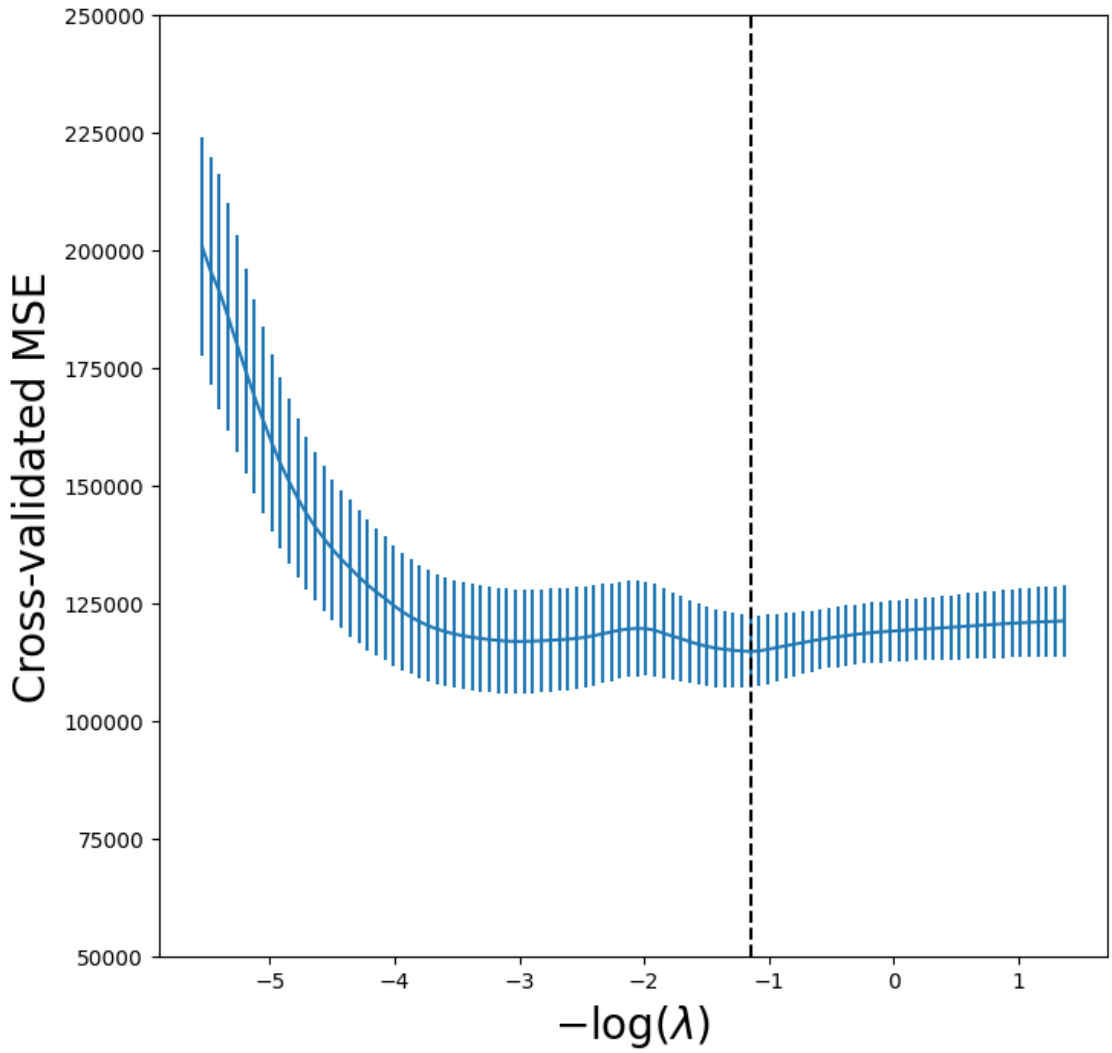
[47]:
```
path_fig, ax = subplots(figsize=(8,8))
soln_path.plot(ax=ax, legend=False)
ax.legend(loc='upper left')
ax.set_xlabel('$-\log(\lambda)$', fontsize=20)
ax.set_ylabel('Standardized coefficiients', fontsize=20);
```



[48]:
```
np.min(tuned_lasso.mse_path_.mean(1))
```

[48]: 114690.73118253502

[49]:
```
lassoCV_fig, ax = subplots(figsize=(8,8))
ax.errorbar(-np.log(tuned_lasso.alphas_),
            tuned_lasso.mse_path_.mean(1),
            yerr=tuned_lasso.mse_path_.std(1) / np.sqrt(K))
ax.axvline(-np.log(tuned_lasso.alpha_), c='k', ls='--')
ax.set_ylim([50000,250000])
ax.set_xlabel('$-\log(\lambda)$', fontsize=20)
ax.set_ylabel('Cross-validated MSE', fontsize=20);
```



[50]:
```
tuned_lasso.coef_
```

[50]:
```
array([-210.01008773,  243.4550306 ,    0.        ,    0.        ,
          0.        ,   97.69397357,  -41.52283116,   -0.        ,
          0.        ,   39.62298193,  205.75273856,  124.55456561,
       -126.29986768,   15.70262427,  -59.50157967,   75.24590036,
         21.62698014,  -12.04423675,   -0.        ])
```

[51]:
```
pca = PCA(n_components=2)
linreg = skl.LinearRegression()
pipe = Pipeline([('pca', pca),
                 ('linreg', linreg)])
pipe.fit(X, Y)
pipe.named_steps['linreg'].coef_
```

[51]: array([0.09846131, 0.4758765])

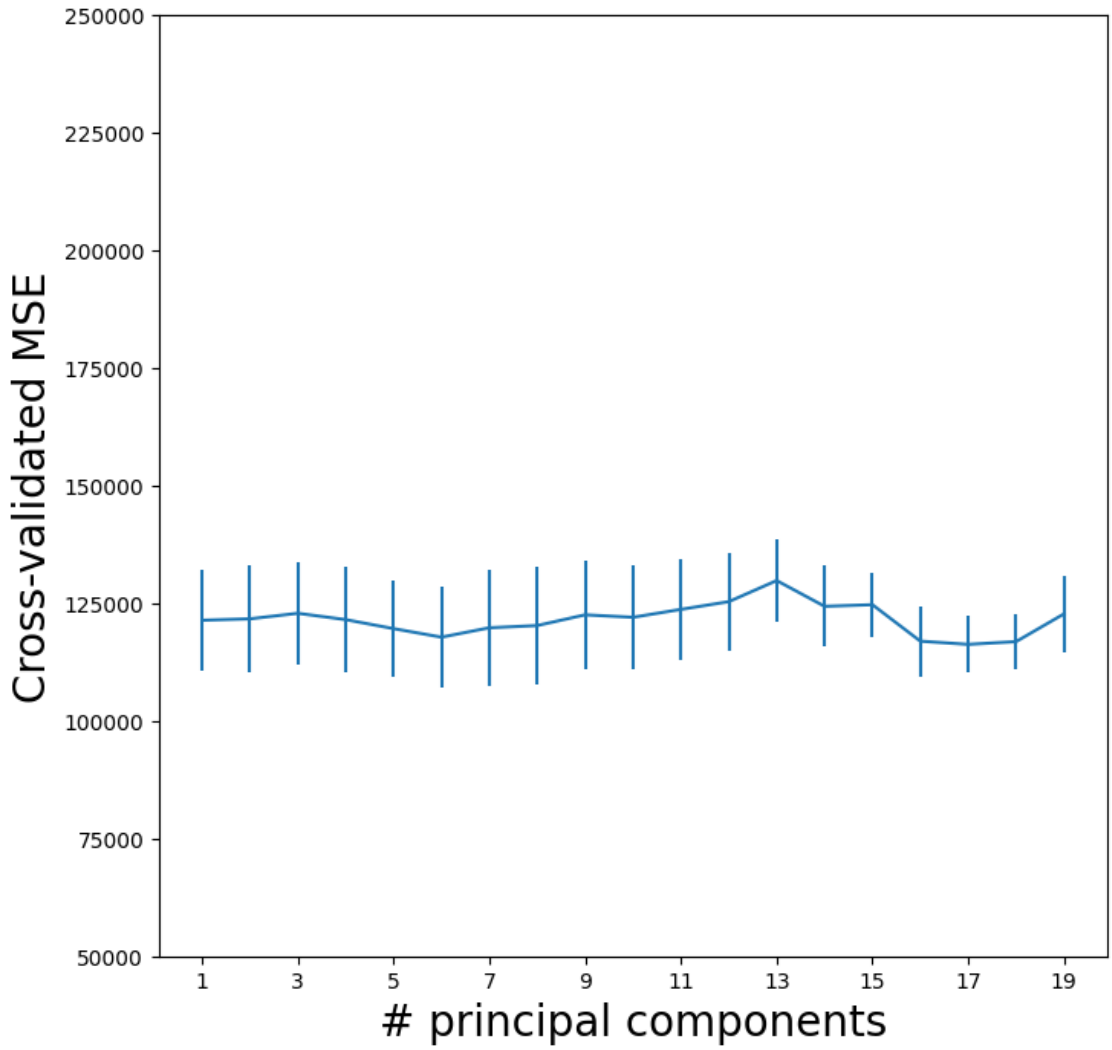
[52]:
```
pipe = Pipeline([('scaler', scaler),
                 ('pca', pca),
                 ('linreg', linreg)])
pipe.fit(X, Y)
pipe.named_steps['linreg'].coef_
```

[52]: array([106.36859204, 21.60350456])

[53]:
```
param_grid = {'pca__n_components': range(1, 20)}
grid = skm.GridSearchCV(pipe,
                        param_grid,
                        cv=kfold,
                        scoring='neg_mean_squared_error')
grid.fit(X, Y)
```

[53]:
```
GridSearchCV(cv=KFold(n_splits=5, random_state=0, shuffle=True),
             estimator=Pipeline(steps=[('scaler', StandardScaler()),
                                       ('pca', PCA(n_components=2)),
                                       ('linreg', LinearRegression())]),
             param_grid={'pca__n_components': range(1, 20)},
             scoring='neg_mean_squared_error')
```

[54]:
```
pcr_fig, ax = subplots(figsize=(8,8))
n_comp = param_grid['pca__n_components']
ax.errorbar(n_comp,
            -grid.cv_results_['mean_test_score'],
            grid.cv_results_['std_test_score'] / np.sqrt(K))
ax.set_ylabel('Cross-validated MSE', fontsize=20)
ax.set_xlabel('# principal components', fontsize=20)
ax.set_xticks(n_comp[::2])
ax.set_ylim([50000,250000]);
```



[55]:
```
Xn = np.zeros((X.shape[0], 1))
cv_null = skm.cross_validate(linreg,
                             Xn,
                             Y,
                             cv=kfold,
                             scoring='neg_mean_squared_error')
-cv_null['test_score'].mean()
```

[55]: 204139.30692994667

[56]:
```
pipe.named_steps['pca'].explained_variance_ratio_
```

[56]: array([0.3831424 , 0.21841076])

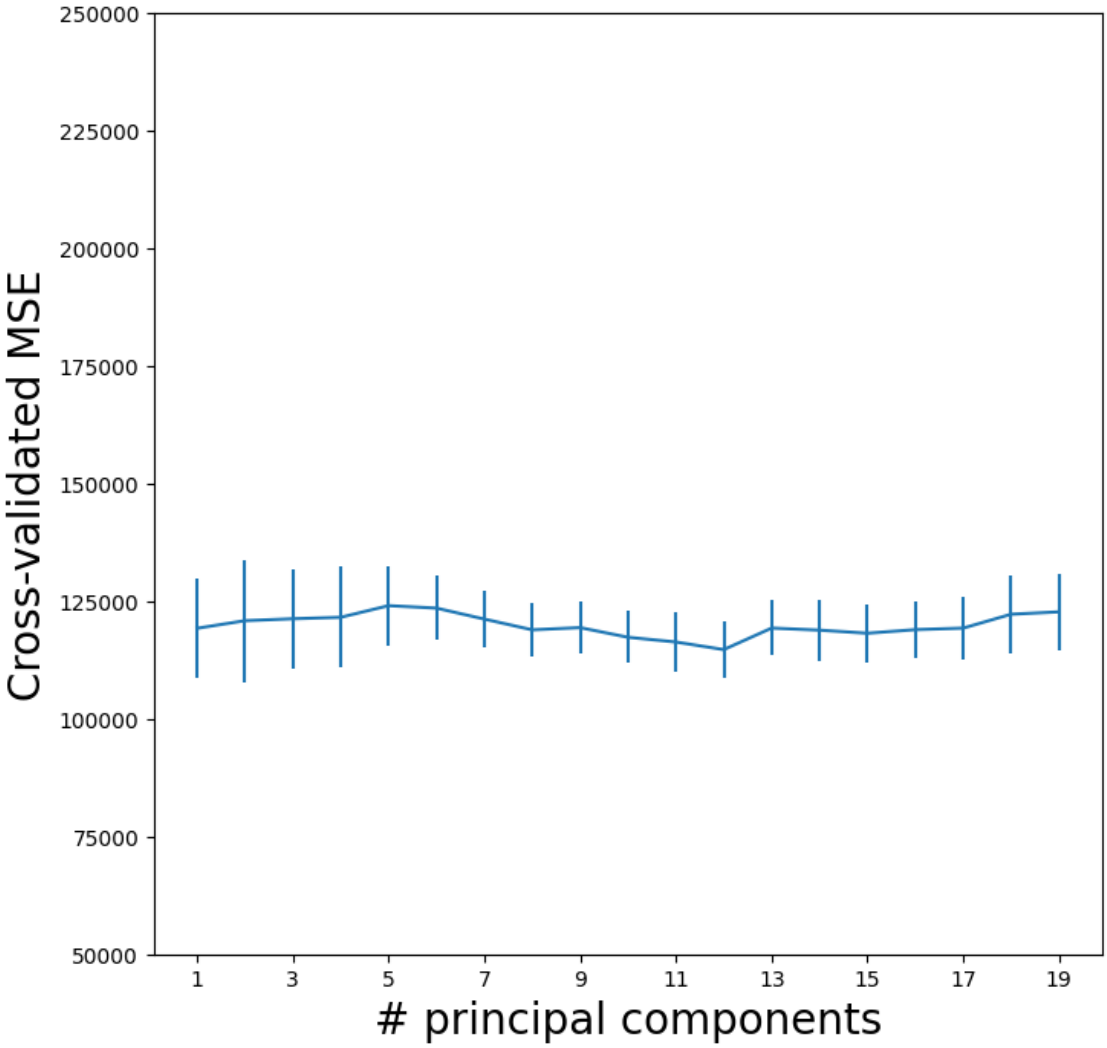
[57]:
```
pls = PLSRegression(n_components=2,
                    scale=True)
pls.fit(X, Y)
```

[57]:
```
PLSRegression()
```

[58]:
```
param_grid = {'n_components':range(1, 20)}
grid = skm.GridSearchCV(pls, param_grid, cv=kfold,␣
 ↪scoring='neg_mean_squared_error')
grid.fit(X, Y)
```

[58]:
```
GridSearchCV(cv=KFold(n_splits=5, random_state=0, shuffle=True),
             estimator=PLSRegression(),
             param_grid={'n_components': range(1, 20)},
             scoring='neg_mean_squared_error')
```

[59]:
```
pls_fig, ax = subplots(figsize=(8,8))
n_comp = param_grid['n_components']
ax.errorbar(n_comp,
            -grid.cv_results_['mean_test_score'],
            grid.cv_results_['std_test_score'] / np.sqrt(K))
ax.set_ylabel('Cross-validated MSE', fontsize=20)
ax.set_xlabel('# principal components', fontsize=20)
ax.set_xticks(n_comp[::2])
ax.set_ylim([50000,250000]);
```


250000
225000
200000
175000
150000
125000
100000
75000
50000
Cross-validated MSE
1
3
5
7
9
11
13
15
17
19
principal components